Cepho System Audit - January 12, 2026

Current State Assessment

1. Database Schema (Complete)

- **58 tables** defined and migrated
- Expert Evolution System: 8 new tables added
  - expert_conversations
  - expert_memory
  - expert_prompt_evolution
  - expert_insights
  - expert_research_tasks
  - expert_collaboration
  - expert_coaching_sessions
  - expert_domain_knowledge

2. Daily Brief System

Status: Schema exists, needs service layer

- `daily_brief_items` table exists with fields:
  - category: key_insight, meeting, task, intelligence, recommendation
  - priority: low, medium, high, critical
  - status: pending, got_it, deferred, delegated, digital_twin

Missing:

- [ ] Daily Brief generation service
- [ ] Morning brief aggregation logic
- [ ] Evening review generation

- [ ] Integration with calendar/tasks/inbox

3. Chief of Staff / Digital Twin

Status: Basic chat implemented, needs orchestration

- Basic chat endpoint exists in routers.ts (lines 123-195)
- System prompt defines Chief of Staff role
- digitalTwinService.ts handles document training

Missing:

- [ ] Chief of Staff orchestration service
- [ ] Expert delegation logic
- [ ] Task prioritization engine
- [ ] Autonomous action capabilities

4. Universal Inbox

Status: UI component exists, needs backend integration

- UniversalInbox.tsx component with drag-drop
- Supports: email, document, voice, image, link, text, whatsapp
- Processing status tracking
- Project creation from selection

Missing:

- [ ] Backend API for inbox items (partially exists)
- [ ] Email integration (OAuth ready, needs connection)
- [ ] WhatsApp integration
- [ ] Auto-categorization with AI

5. Expert Evolution System

Status: Just built - database + API complete

- Database schema: Complete
- API endpoints: Complete (expertEvolution router)
- ExpertPromptBuilder service: Complete
- Memory extraction: Complete

Missing:

- [ ] Wire into actual expert chat flow
- [ ] Auto-save conversations after each interaction
- [ ] Auto-extract memories from conversations
- [ ] Chief of Staff coaching loop

6. AI Expert Engine

Status: 287 experts defined, needs evolution wiring

- aiExperts.ts with full roster
- Expert categories and composites defined
- Performance tracking schema exists

Missing:

- [ ] Wire expertPromptBuilder into expert calls
- [ ] Inject memory context into prompts
- [ ] Store conversations after each interaction

7. Integrations

Status: OAuth infrastructure ready

- Integration table in database
- OAuth flow components exist
- Supported: Asana, Outlook, Gmail, Google Calendar

Missing:

- [ ] Actual OAuth credentials configured

- [ ] Webhook handlers for real-time updates
- [ ] Data sync services

Priority Actions for Completion

Phase 1: Wire Expert Evolution (Critical)

1. Modify expert chat to use expertPromptBuilder
2. Auto-save conversations after each expert interaction
3. Auto-extract memories via Chief of Staff

Phase 2: Chief of Staff Orchestration

1. Create chiefOfStaffService.ts
2. Implement task delegation to experts
3. Build daily brief generation
4. Evening review aggregation

Phase 3: Integration Flows

1. Connect inbox to backend API
2. Email forwarding setup
3. Calendar sync

Files to Create/Modify

1. `/server/services/chiefOfStaffService.ts` - NEW
2. `/server/services/dailyBriefService.ts` - NEW
3. `/server/services/inboxProcessingService.ts` - NEW
4. `/server/routers.ts` - Add daily brief routes
5. `/client/src/hooks/useExpertChat.ts` - Wire evolution